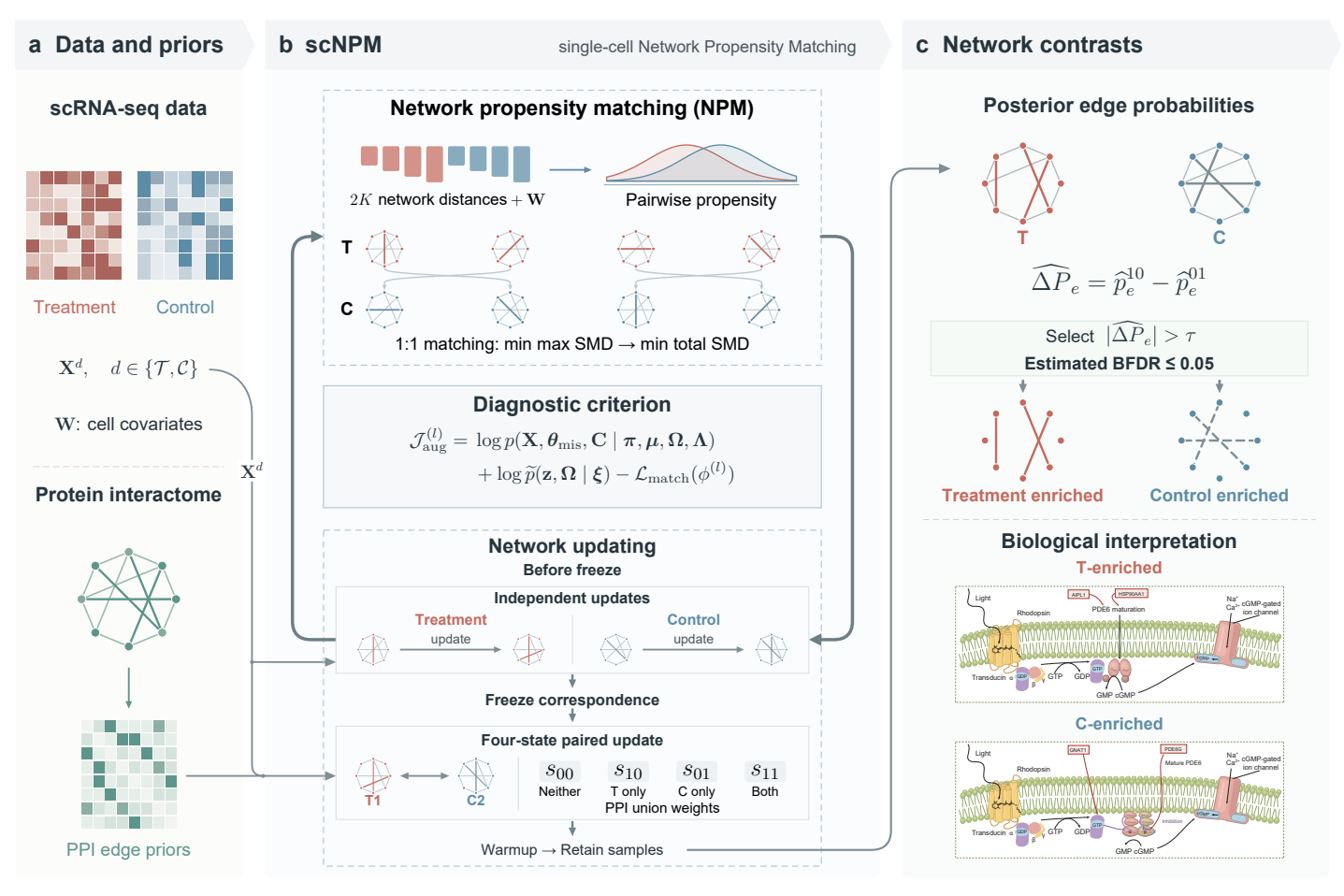

a Data and priors
b scNPM
single-cell Network Propensity Matching
c Network contrasts
scRNA-seq data
Treatment
Control
$\mathbf{X}^d, \quad d \in \{\mathcal{T}, \mathcal{C}\}$
$\mathbf{W}$: cell covariates
Protein interactome
PPI edge priors
$\mathbf{X}^d$
Network propensity matching (NPM)
$2K$ network distances + $\mathbf{W}$
Pairwise propensity
T
C
1:1 matching: min max SMD → min total SMD
Diagnostic criterion
$\mathcal{J}_{\text{aug}}^{(l)} = \log p(\mathbf{X}, \boldsymbol{\theta}_{\text{mis}}, \mathbf{C} \mid \boldsymbol{\pi}, \boldsymbol{\mu}, \boldsymbol{\Omega}, \boldsymbol{\Lambda})$
$+ \log \widetilde{p}(\mathbf{z}, \boldsymbol{\Omega} \mid \boldsymbol{\xi}) - \mathcal{L}_{\text{match}}(\phi^{(l)})$
Network updating
Before freeze
Independent updates
Treatment
update
Control
update
Freeze correspondence
Four-state paired update
T1
C2
s_{00} Neither
s_{10} T only
s_{01} C only
s_{11} Both
PPI union weights
Warmup → Retain samples
Posterior edge probabilities
T
C
$\widehat{\Delta P}_e = \widehat{p}_e^{10} - \widehat{p}_e^{01}$
Select $|\widehat{\Delta P}_e| > \tau$
Estimated BFDR ≤ 0.05
Treatment enriched
Control enriched
Biological interpretation
T-enriched
C-enriched
Light
Rhodopsin
PDE6 maturation
Transducin α
GTP
GDP
GMP cGMP
Na+ Ca2+ cGMP-gated ion channel
Mature PDE6
inhibition